

# Noise spectroscopy through dynamical decoupling with a superconducting flux qubit

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**Quantum coherence in natural and artificial spin systems is fundamental to applications ranging from quantum information science to magnetic-resonance imaging and identification. Several multipulse control sequences targeting generalized noise models have been developed to extend coherence by dynamically decoupling a spin system from its noisy environment. In any particular implementation, however, the efficacy of these methods is sensitive to the specific frequency distribution of the noise, suggesting that these same pulse sequences could also be used to probe the noise spectrum directly. Here we demonstrate noise spectroscopy by means of dynamical decoupling using a superconducting qubit with energy-relaxation time  $T_1 = 12 \mu\text{s}$ . We first demonstrate that dynamical decoupling improves the coherence time  $T_2$  in this system up to the  $T_2 = 2 T_1$  limit (pure dephasing times exceeding  $100 \mu\text{s}$ ), and then leverage its filtering properties to probe the environmental noise over a frequency ( $f$ ) range 0.2–20 MHz, observing a  $1/f^\alpha$  distribution with  $\alpha < 1$ . The characterization of environmental noise has broad utility for spin-resonance applications, enabling the design of optimized coherent-control methods, promoting device and materials engineering, and generally improving coherence.**

Decoherence of a quantum superposition state in natural and artificial two-level spin systems arises from the interaction between the constituent system and the uncontrolled degrees of freedom in its environment. Within the standard Bloch–Redfield picture of two-level system dynamics, decoherence is characterized by two rates: a longitudinal relaxation rate  $\Gamma_1 = 1/T_1$  due to the exchange of energy with the environment, and a transverse relaxation rate  $\Gamma_2 = \Gamma_1/2 + \Gamma_\phi$ , which contains the pure dephasing rate  $\Gamma_\phi$ . Irreversible energy relaxation can only be mitigated by reducing the amount of environmental noise, by reducing the qubit’s internal sensitivity to that noise, or through multiqubit encoding and error-correction protocols (which already presume ultralow error rates). In contrast, dephasing is in principle reversible and can be refocused dynamically through the application of coherent control-pulse methods.

Since the 1950s, several single-<sup>1</sup> and multipulse<sup>2–4</sup> sequences have been developed within the field of nuclear magnetic resonance<sup>4</sup> and later extended to electron spin resonance. More recently, such dynamical-decoupling techniques have been applied to mitigate noise and extend coherence in qubits based on atomic ensembles<sup>5–7</sup> and single atoms<sup>8</sup>, spin ensembles<sup>9</sup>, semiconductor quantum dots<sup>10,11</sup> and diamond nitrogen–vacancy centres<sup>12,13</sup>. In most experimental<sup>5–13</sup> and theoretical<sup>14–20</sup> dynamical-decoupling works, the focus has been on using pulse sequences to minimize errors (that is, extend  $T_2$ ) within an operating time  $\tau$  such that  $\tau \ll T_2$ , for example, to reach fault-tolerant error thresholds.

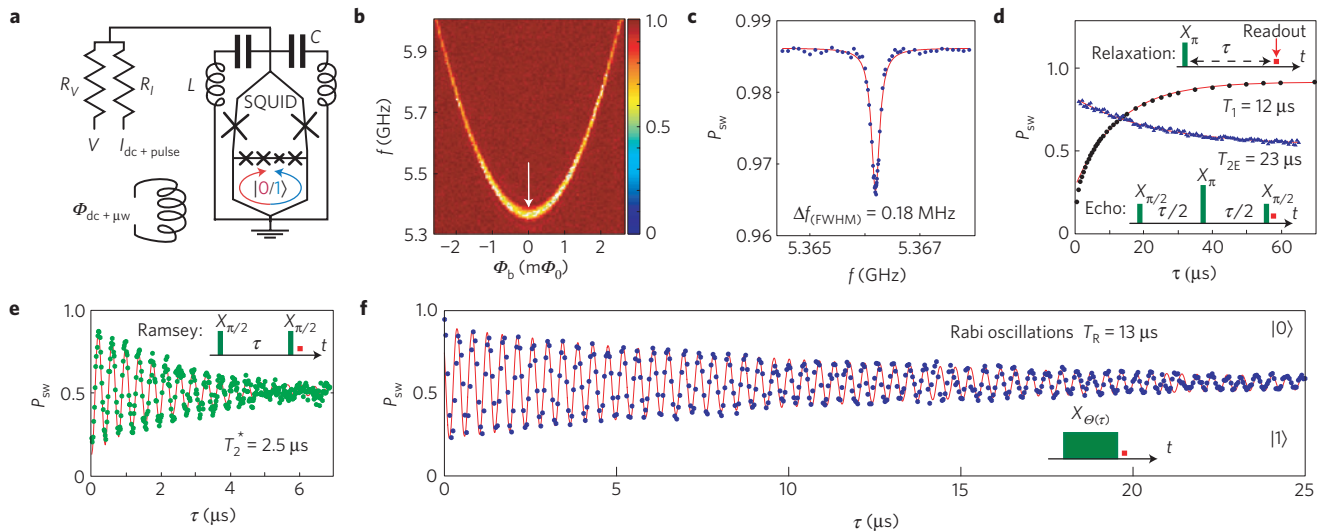
We extend multipulse dynamical decoupling to superconducting qubits<sup>21</sup>. However, our interest here is more general. Our goal is to measure the environmental noise spectrum<sup>19,22</sup>, which

we accomplish by monitoring the response of the spin to its environment, as mediated by dynamical-decoupling pulses, over a range of times that extends from the small-error limit ( $\tau \ll T_2$ ) to the large-error limit ( $\tau \sim T_2$ ). This should not be viewed as unique to our purpose; indeed, applications such as magnetic resonance imaging and sensing generally operate in the presence of large errors with variability from sample to sample such that there is probably no generally optimal pulse sequence<sup>23</sup>. In these cases, characterizing the noise-power spectral density (PSD) precisely would greatly facilitate the design of tailored pulse sequences<sup>4,14</sup> that can reduce error rates, reduce the pulse number or power required to achieve a given error rate and otherwise engineer an improved measurement outcome.

In our test system, a superconducting (persistent-current) flux qubit<sup>24,25</sup>, we carry out noise spectroscopy by dynamical decoupling over the wide frequency range 0.2–20 MHz: the obtained noise PSD is a  $1/f^\alpha$  power-law with  $\alpha \approx 0.9$ . We independently confirm this noise PSD with that obtained from a Rabi-spectroscopy approach and, for completeness, characterize the high-frequency PSD over the range 5.4–21 GHz by monitoring the qubit’s energy-relaxation rate<sup>26</sup>.

The device (Fig. 1a, Supplementary Fig. S1) is an aluminium loop interrupted by four Al–AlO<sub>x</sub>–Al Josephson junctions. When an external magnetic flux  $\Phi$  threading the loop is close to half a superconducting flux quantum  $\Phi_0/2$ , the diabatic states correspond to clockwise and anticlockwise persistent currents  $I_p = 0.18 \mu\text{A}$  with energies  $\pm \hbar \epsilon / 2 = \pm I_p \Phi_b$ , tunable by the flux bias  $\Phi_b = \Phi - \Phi_0/2$ . At  $\Phi_b = 0$ , the degenerate persistent-current states hybridize with a strength  $\hbar \Delta = \hbar \times 5.3662 \text{ GHz}$  (Fig. 1b, Supplementary Fig. S2b),

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**Figure 1 | Qubit device and characterization.** **a**, Device and biasing schematic representation: an aluminium superconducting loop interrupted by Josephson junctions (crosses) with a readout d.c. SQUID. **b**, Frequency spectroscopy of the qubit's  $|0\rangle \rightarrow |1\rangle$  transition. **c**, Spectroscopy at  $\Phi_b = 0$  (arrow in **b**). **d**, Echo decay (blue triangles) and relaxation from the excited state (black circles) at  $\Phi_b = 0$ . In the insets,  $\tau$  is a time delay and  $X_\Theta$  symbolizes a rotation of the Bloch vector by the angle  $\Theta$  around the axis  $\hat{\sigma}_x$ . The red squares indicate the readout. **e**, Free-induction decay (Ramsey fringe) at  $\Phi_b = 0$ . **f**, Rabi oscillations at  $\Phi_b = 0$ .

where  $\hbar = h/2\pi$  and  $h$  is Planck's constant. The corresponding two-level Hamiltonian is

$$\hat{\mathcal{H}} = -\frac{\hbar}{2}[(\varepsilon + \delta\varepsilon)\hat{\sigma}_x + (\Delta + \delta\Delta)\hat{\sigma}_z] \quad (1)$$

which includes noise fluctuation terms  $\delta\varepsilon$  and  $\delta\Delta$ , and  $\hat{\sigma}_{x,z}$  are Pauli operators. The ground ( $|0\rangle$ ) and excited ( $|1\rangle$ ) states have frequency splitting  $\omega_{01} = \sqrt{\varepsilon^2 + \Delta^2}$  and are well isolated owing to the qubit's large anharmonicity,  $\omega_{12}/\omega_{01} \approx 5$ . Noise and decoherence in superconducting qubits have been studied theoretically<sup>27–30</sup> and experimentally<sup>26,29,31</sup>. In our qubit, the environmental noise leading to fluctuations  $\delta\varepsilon$  (for example, flux noise) and  $\delta\Delta$  (for example, critical current and charge noise) physically couples to the qubit in the  $\varepsilon$ – $\Delta$  frame (equation (1)). However, their manifestation as longitudinal noise (dephasing) or transverse noise (energy relaxation) is tunable<sup>29</sup> by the flux bias  $\Phi_b$  and determined, respectively, by their projections  $\delta\omega_{z'}$  onto the qubit's quantization axis  $\hat{\sigma}_{z'}$  (which makes an angle  $\theta = \arctan(\varepsilon/\Delta)$  with  $\hat{\sigma}_z$ ) and  $\delta\omega_{\perp'}$  onto the plane perpendicular to  $\hat{\sigma}_{z'}$ .

The chip is mounted in a  $^3\text{He}/^4\text{He}$  dilution refrigerator with 12 mK base temperature. For each experimental trial, we initialize the qubit by waiting sufficient time ( $\sim 1$  ms) for it to relax to its ground state. We drive the desired quantum-state rotations of angle  $\Theta$  by applying calibrated in-phase ( $X_\Theta$ ) and quadrature ( $Y_\Theta$ ) harmonic flux pulses to the qubit loop. The pulses consist of Gaussian envelopes with a typical standard deviation  $\sigma = 1.2$  ns and truncated at  $\pm 3\sigma$ . The qubit readout has 79% visibility (Supplementary Fig. S3) and is carried out in the energy basis by determining the switching probability  $P_{sw}$  of a hysteretic dc SQUID, averaging over several thousand trials (see Supplementary Information C).

We begin with a spectroscopic characterization of our device in the absence of multipulse dynamical decoupling. The qubit level splitting  $\omega_{01}$  is measured through saturated frequency spectroscopy (Fig. 1b), and at low power it exhibits a Lorentzian full-width-at-half-maximum linewidth  $\Delta f_{\text{FWHM}} = 0.18$  MHz at  $\Phi_b = 0$  (Fig. 1c). The energy relaxation is generally exponential and its time constant  $T_1 = 12 \pm 1$   $\mu$ s (Fig. 1d) is remarkably long among superconducting

qubits<sup>21</sup>, about six times longer than a similarly designed and fabricated device<sup>31</sup>. We observe very long decay times at  $\Phi_b = 0$  for the Hahn spin echo,  $T_{2E} = 23$   $\mu$ s (Fig. 1d), Ramsey free induction,  $T_2^* = 2.5$   $\mu$ s (Fig. 1e) and Rabi oscillations,  $T_R = 13$   $\mu$ s (Fig. 1f). The dephasing times decrease (rates increase) dramatically away from  $\Phi_b = 0$  owing to the qubit's increased sensitivity  $|\partial\omega_{01}/\partial\varepsilon|$  to the dominant  $\delta\varepsilon$ -noise (flux noise) in this system<sup>31</sup>. Furthermore, although the spin echo and Rabi oscillations exhibit an essentially  $T_1$ -limited, exponential decay at  $\Phi_b = 0$ , in general their noise-dependent decay functions are non-exponential, indicating noise sources with long correlation times, singular near  $\omega \approx 0$  (for example,  $1/f$ -type noise at low frequencies, relevant to dephasing).

The frequency distribution of the noise power for a noise source  $\lambda$  that leads to decoherence is characterized by its PSD  $S_\lambda(\omega) = (1/2\pi) \int_{-\infty}^{\infty} dt \langle \lambda(0)\lambda(t) \rangle \exp(-i\omega t)$ . A superposition state's accumulated phase  $\varphi(t) = \langle \omega_{01} \rangle t + \delta\varphi(t)$  diffuses owing to adiabatic fluctuations of the transition frequency,  $\delta\varphi(t) = (\partial\omega_{01}/\partial\lambda) \int_0^t dt' \delta\lambda(t')$ , where  $\partial\omega_{01}/\partial\lambda$  is the qubit's longitudinal sensitivity to  $\lambda$ -noise. For noise generated by a large number of fluctuators that are weakly coupled to the qubit, the statistics are Gaussian. Ensemble averaging over all realizations of the stochastic process  $\delta\lambda(t)$ , the dephasing is  $\langle \exp[i\delta\varphi(t)] \rangle \equiv \exp[-\chi_N(t)]$ , with the coherence integral

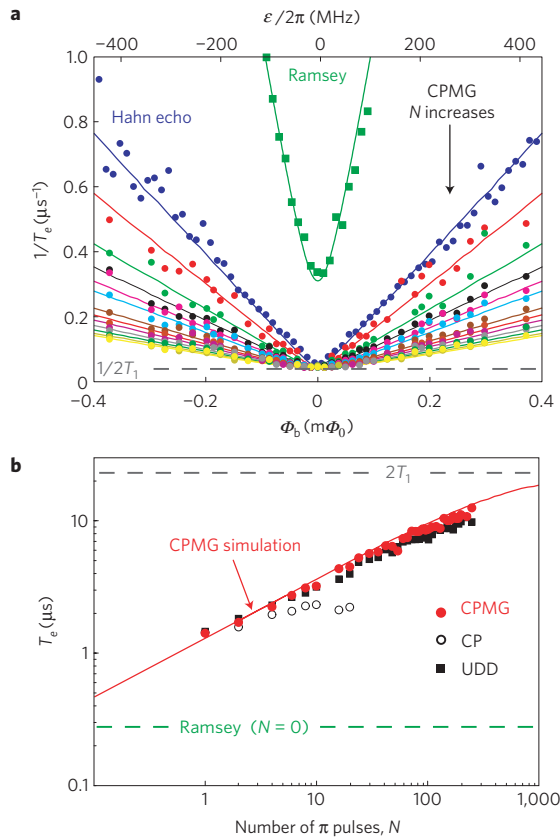
$$\chi_N(\tau) = \tau^2 \sum_{\lambda} \left( \frac{\partial\omega_{01}}{\partial\lambda} \right)^2 \int_0^\infty d\omega S_\lambda(\omega) g_N(\omega, \tau) \quad (2)$$

where  $\tau$  is the free evolution time,  $N$  will denote the number of  $\pi$ -pulses in the pulse sequences<sup>5,17</sup> and  $g_N$  is a dimensionless weighting function defined in equation (3).

The function  $g_N(\omega, \tau)$  can be viewed as a frequency-domain filter of the noise  $S_\lambda(\omega)$ . Its filter properties depend on the number  $N$  and distribution of  $\pi$ -pulses<sup>5,6,17,19,32</sup>,

$$g_N(\omega, \tau) = \frac{1}{(\omega\tau)^2} \left| 1 + (-1)^{1+N} \exp(i\omega\tau) + 2 \sum_{j=1}^N (-1)^j \exp(i\omega\delta_j\tau) \cos(\omega\tau_j/2) \right|^2 \quad (3)$$





**Figure 3 | Dephasing under the CPMG sequence.** **a**, Decay rates (inverse of  $1/e$  times) versus flux detuning: free induction (Ramsey,  $N=0$ , green squares) and CPMG (coloured circles) with  $N=1, 2, 4, 6, 8, 10, 16, 20, 24, 30, 36, 42, 48$  (colours correspond to those in Fig. 2c). Solid lines are calculations using equation (2) with parameters in Table 1. **b**,  $1/e$  decay time under  $N$ -pulse CPMG, CP and UDD sequences at  $\Phi_b = -0.4 m\Phi_0$  ( $\varepsilon/2\pi = 430$  MHz). The simulation (red line) assumes ideal pulses and noise acting during the free evolution.

observe a coherence improvement going from  $N=1$  to  $N=2$   $\pi$ -pulses beyond  $\varepsilon/2\pi = 6.4$  GHz, corresponding to a qubit frequency of  $\omega_{01}/2\pi > 8.4$  GHz (see Supplementary Fig. S5). Although the coherence times generally decrease as  $|\varepsilon|$  is increased owing to the qubit's increased sensitivity to flux noise, the fractional improvement in coherence is essentially constant over the entire

range. The echo-decay time  $T_{2E}$  remains considerably longer than the  $\pi$ -pulse time  $\tau_\pi$ , indicating that the refocusing remains efficient.

At a specific flux bias  $\Phi_b = -0.4 m\Phi_0$ , where the qubit is highly sensitive to  $\delta\varepsilon$  noise, the CPMG sequence gives a marked improvement in the decay time  $T_e$  up to  $N \approx 200$   $\pi$ -pulses, beyond which pulse errors begin to limit the CPMG efficiency (Fig. 3b). We achieve a 50-fold enhancement of  $T_2^{\text{CPMG}}$  over the Ramsey  $T_2^*$ , and a well over 100-fold improvement in the Gaussian pure dephasing time  $T_\varphi$ . The CPMG sequence performs about 5% better than UDD, indicating that the  $1/f$   $\delta\varepsilon$ -noise spectrum exhibits a relatively soft (if any) high-frequency cutoff<sup>19,20</sup>, because the UDD filter function has larger side-lobes at higher frequencies (Fig. 2c). CPMG also dramatically outperforms CP, despite having the same filter function, as  $Y_\pi$ -pulse errors appear only to fourth order with CPMG, whereas with CP  $X_\pi$  errors accumulate to second order<sup>33</sup>.

Given its superior performance mitigating noise, we in turn use the filtering property of the CPMG sequence to characterize the flux-noise ( $\delta\varepsilon$ ) spectrum. The filter  $g_N(\omega, \tau)$  is sufficiently narrow about  $\omega'$  that we can treat the noise as constant within its bandwidth  $B$  and approximate equation (2) as  $\chi_N(\tau) \approx \tau^2 (\partial\omega_{01}/\partial\varepsilon)^2 S_\varepsilon(\omega') g_N(\omega', \tau) B$  (Fig. 2e). We compute  $\omega'$  and  $B$  numerically for each  $N$  and  $\tau$  used in the CPMG measurements of Fig. 3b. The measured decay function contains three decay rates: dephasing  $\Gamma_\varphi^{(N)}$  and exponential relaxation  $\Gamma_1/2$  during the total free-evolution time  $\tau$ , and pulse-induced decay  $\Gamma_p$  during  $N\tau_\pi$ . Using a recursive method explained in detail in part I of Supplementary Information, we effectively divide out the  $\Gamma_1$  and  $\Gamma_p$  components from the raw data and compute the noise PSD  $S_\varepsilon(\omega)$  without presuming a functional form for the decay functions or the spectrum. This procedure yields a  $1/f^\alpha$ -type PSD  $S_\varepsilon(\omega)$  over the region 0.2–20 MHz in Fig. 5, with a slight increase in the measured PSD above 2 MHz. Interestingly, by fitting the lower-frequency, linear portion we find that the PSD is approximated by a  $1/f^\alpha$  power law<sup>34</sup> with  $\alpha \approx 0.9$  (solid, red line) and noise amplitude  $A_\varphi = (0.8 \mu\Phi_0)^2$ .

To check the noise-spectroscopy results we analysed the decoherence during driven evolution<sup>29,35</sup> (see Supplementary Information J for details). A transverse driving field at frequency  $\omega$  and drive strength  $\Omega$  results in Rabi oscillations with angular nutation frequency  $\Omega_R = \sqrt{\Omega^2 + (\Delta\omega)^2} \approx \Omega + (\Delta\omega)^2/2\Omega$ , where  $\Delta\omega = \omega - \omega_{01}$ . The noise at the Rabi frequency,  $\Gamma_\Omega^{(\lambda)} = \pi S_\lambda(\Omega_R)$ , along with  $T_1$  relaxation, comprises the usual exponential Rabi-decay rate

$$\Gamma_R = \left( \frac{3}{4} \Gamma_1 + \frac{1}{2} \Gamma_\Omega^{(\Delta)} \right) \cos^2 \theta + \left( \frac{\varepsilon}{\omega_{01}} \right)^2 \frac{1}{2} \Gamma_\Omega^{(\varepsilon)} \quad (4)$$

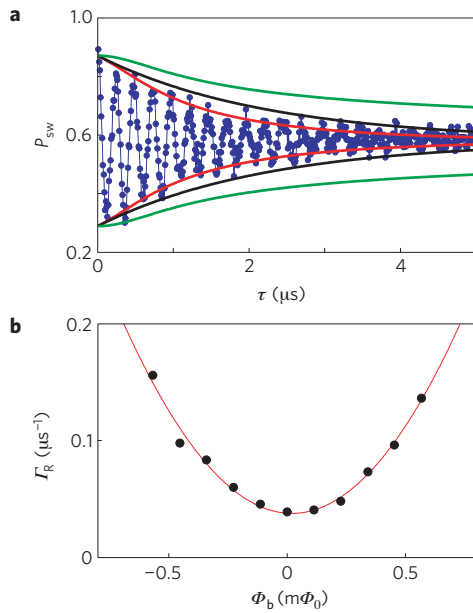
**Table 1 | Coherence times and quasi-static noise parameters.**

| Noise parameters                                               | $\sigma_\lambda/2\pi$ | $\omega_{\text{low}}^\lambda/2\pi$ | $\omega_{\text{high}}^\lambda/2\pi$ | $A_{\varphi, i_c}$              |
|----------------------------------------------------------------|-----------------------|------------------------------------|-------------------------------------|---------------------------------|
| $\lambda = \varepsilon$ (equivalent $\Phi$ noise)              | 10 MHz                | 1 Hz                               | 1 MHz                               | $(1.7 \times 10^{-6} \Phi_0)^2$ |
| $\lambda = \Delta$ (equivalent $i_c$ noise)                    | 0.06 MHz              | 1 Hz                               | < 0.1 MHz                           | $(4.0 \times 10^{-6})^2$        |
| Coherence times                                                | $T_1$                 | $T_2^*$                            | $T_2^{\text{CPMG}}$                 | $T_2^{\text{CPMG}}/T_2^*$       |
| $\Phi_b = 0 m\Phi_0$ ( $\delta\Delta$ noise dominates)         | 12 $\mu\text{s}$      | 2.5 $\mu\text{s}$                  | 23 $\mu\text{s}$ ( $N=1$ )          | 9                               |
| $\Phi_b = -0.4 m\Phi_0$ ( $\delta\varepsilon$ noise dominates) | 12 $\mu\text{s}$      | 0.27 $\mu\text{s}$                 | 13 $\mu\text{s}$ ( $N=200$ )        | 48                              |

Coherence times and their improvement under dynamical decoupling at flux biases  $\Phi_b$  where  $\delta\varepsilon$  and  $\delta\Delta$  noises dominate. Assuming that flux ( $\Phi$ ) and normalized critical current ( $i_c = \delta i_c / i_c$ ) are responsible for all  $\delta\varepsilon$  and  $\delta\Delta$  noise, respectively, we convert these noises to frequency noise by the sensitivities  $\kappa_\varepsilon = \partial\varepsilon/\partial\Phi = 6.9 \times 10^9 \text{ s}^{-1}/m\Phi_0$  and  $\kappa_\Delta = \partial\Delta/\partial i_c = 20 \times 10^9 \text{ s}^{-1}$  (see Supplementary Information D). We then describe the quasi-static, Gaussian noise distributions by their standard deviations,  $\sigma_\lambda$ , obtained by integrating the noises  $S_{\varepsilon, \Delta}(\omega) = \kappa_{\varepsilon, \Delta}^2 A_{\varphi, i_c} \omega$  over the bandwidth given by the experimental protocol:

$$\sigma_\lambda^2 = 2 \int_{\omega_{\text{low}}^\lambda}^{\omega_{\text{high}}^\lambda} d\omega S_\lambda(\omega).$$

The cutoff  $\omega_{\text{low}}^\lambda$  is given by the averaging time over all trials, typically 1 s for 1,000 averages at 1 kHz repetition rate, and  $\omega_{\text{high}}^\lambda$  by the typical free-evolution time,  $\tau = 1 \mu\text{s}$ , during a single trial. The noise strengths,  $A_{\varphi, i_c}$ , are derived from the Ramsey and echo data, assuming a power law  $1/f^{\alpha=1}$ . These agree with previously reported values<sup>31,39</sup>. At  $\varepsilon = 0$ , the dephasing improvement under a Hahn echo is greater than the theory would suggest; the lower  $\omega_{\text{high}}^\Delta$  gives consistency. Simulations using these parameters yield agreement with the Ramsey-fringe and Hahn-echo data (see Supplementary Information F), and they are consistent with equation (2).



**Figure 4 | Decoherence during driven dynamics.** **a**, Rabi oscillations with  $\Omega_R/2\pi = 2$  MHz at  $\varepsilon/2\pi = 225$  MHz. The red line envelope is a fitting using  $\zeta(t)$  and  $\Gamma_R$ . The black line shows the  $\Gamma_R$  decay only, and the green line the  $\zeta(t)$ -envelope contribution. **b**, Rabi-decay rate  $\Gamma_R$  versus flux detuning at  $\Omega_R/2\pi = 2$  MHz, with a parabolic fit to equation (4) used to obtain  $S_\varepsilon(\Omega) = (1/\pi) \Gamma_\Omega^{(\varepsilon)}$ .

where  $\cos^2\theta \approx 1$  as the quantization angle is small. In addition, accounting for the quasi-static noise with variance  $\sigma_\varepsilon^2$  (Table 1), we obtain the decay function  $\zeta(\tau) = (1 + (u\tau)^2)^{-1/4}$ , where  $u = (\varepsilon/\omega_0)^2 \sigma_\varepsilon^2/\Omega$ . Figure 4a shows the combined decay function  $\zeta(\tau) \times \exp(-\Gamma_R\tau)$ .

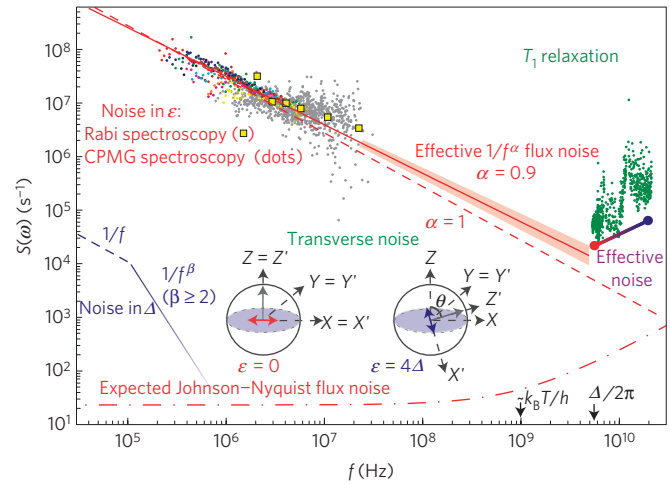
To determine  $S_\lambda(\Omega_R)$ , we measured the Rabi oscillations versus  $\Phi_b$  with fixed Rabi frequency  $\Omega_R$ . For each  $\Omega_R$ , we find the  $\varepsilon$ -independent part of the rate by fitting the envelope of the oscillations at  $\varepsilon = 0$ . The rate  $\Gamma_\Omega^{(\Delta)}$  was too small to distinguish accurately from  $\Gamma_1$ , consistent with its correspondingly small quasi-static noise  $\sigma_\Delta^2$  (Table 1). Then, for  $\varepsilon \neq 0$ , we divide out the known quasi-static contribution  $\zeta(\tau)$  and fit to the parabolic term in equation (4), from which we obtain  $\Gamma_\Omega^{(\varepsilon)}$  (Fig. 4b). Using this approach, we find  $S_\varepsilon(\Omega)$  to be consistent with the  $1/f^\alpha$  noise obtained from the CPMG measurements (Fig. 5).

For completeness, we now turn to transverse noise at the qubit frequency  $\omega_0$  responsible for energy relaxation  $\Gamma_1$ , that is,  $\delta\varepsilon$ -noise at  $\varepsilon = 0$  and  $\delta\Delta$ -noise at  $\varepsilon \gg \Delta$  (Fig. 5, inset). In the low-temperature limit,  $k_B T \ll \hbar\omega_0$ , where the environment cannot excite the qubit, the golden-rule expression for  $\Gamma_1$  in a weakly damped quantum two-level system is

$$\Gamma_1 = \frac{\pi}{2} \sum_{\lambda \in \delta\varepsilon, \delta\Delta} \left( \frac{\partial \omega_{\perp}}{\partial \lambda} \right)^2 S_\lambda(\omega_0) = \frac{\pi}{2} S(\omega_0) \quad (5)$$

with  $\partial \omega_{\perp}/\partial \lambda$  the qubit's sensitivity to transverse noise and  $S(\omega_0)$  the total PSD (see Supplementary Information K).

We apply a long ( $\gg T_1$ ) microwave pulse to saturate the transition and monitor the energy decay to the ground state, using equation (5) to determine  $S(\omega_0)$  over the frequency range  $\Delta \leq \omega_0 \leq 2\pi \times 21$  GHz by tuning  $\Phi_b$ . At  $\Phi_b = 0$  ( $\omega_0 = \Delta$ ) and using a measurement-repetition period  $t_{\text{rep}} > 1$  ms, we observe  $T_1 = 12 \pm 1$   $\mu$ s as shown in Fig. 1d. As  $t_{\text{rep}}$  becomes shorter than 1 ms, the decay becomes increasingly non-exponential, which we attribute to the residual presence of non-equilibrium quasiparticles



**Figure 5 | Noise-power spectral density (PSD).** Multicoloured dots,  $\delta\varepsilon$ -noise PSD (0.2–20 MHz) derived from CPMG data at  $\Phi_b = -0.4$   $m\Phi_0$  (see the text). Colours correspond to the various  $N$  in Fig. 3a; grey dots for data up to  $N = 250$ . Yellow squares,  $\delta\varepsilon$ -noise PSD (2–20 MHz) derived from Rabi spectroscopy (Fig. 4b and Supplementary Information J). Diagonal, dashed lines, Estimated flux (red) and  $\delta\Delta$  (blue)  $1/f$  noises, converted to frequency, inferred from the Ramsey and echo measurements (see  $A_{\Phi_{ic}}$  and  $\omega_{\text{high}}^\Delta$  in Table 1). Solid, red line, Power-law dependence,  $S_\varepsilon(2\pi f) = \kappa_\varepsilon^2 A_\Phi / (2\pi f)^\alpha$ , extrapolated beyond the qubit's frequency,  $\Delta/2\pi$ . Fitting the low-frequency, linear portion of the measured PSD derived from CPMG data (before the slight upturn beyond 2 MHz) yields the parameters  $A_\Phi = (0.8 \mu\Phi_0)^2$  and  $\alpha = 0.9$  (see Supplementary Information I.3). The shaded area covers  $\alpha \pm 0.05$ . Green dots, High-frequency transverse noise is purely  $\delta\varepsilon$ -noise at  $\Delta/2\pi = 5.4$  GHz and becomes predominantly  $\delta\Delta$ -noise at higher frequency. Purple line, Guide to indicate linearly increasing Nyquist (quantum) noise, including the eigenbasis rotation (see the inset); circles indicate transverse  $\delta\varepsilon$ - (red) and  $\delta\Delta$ - (blue) noises. Dash-dotted line, Expected thermal and quantum noise (equation (6)). Inset, Graphic representation of the quantization axis (grey arrows of fixed length) with the qubit's ( $Z'$ ) eigenstate tilted from the 'laboratory' frame ( $Z$ ) by the angle  $\theta$ . Fields  $\varepsilon(\Phi_b)$  and  $\Delta$  point in the  $X$  and  $Z$  directions, respectively. Red and blue double arrows indicate transverse noise.

generated by the switching SQUID during readout. We observe structure in the  $\Gamma_1$ -data due to environmental modes (for example cavity modes, impedance resonances) with uncontrolled couplings to the qubit (Fig. 5). For comparison, we plot the expected Johnson–Nyquist flux noise in Fig. 5 due to the  $G = 1/50 \Omega$  environment mutually coupled with strength  $M = 0.02$  pH to the qubit through the microwave line,

$$S_\varepsilon^{\text{JN}}(\omega) = \frac{1}{2\pi} \left( \frac{\partial \varepsilon}{\partial \Phi} \right)^2 M^2 \frac{2\hbar\omega G}{1 - e^{-\hbar\omega/k_B T}} \quad (6)$$

This known noise source falls about 100 times below the measured PSD at  $f_0 = \Delta/2\pi$  (red circle in Fig. 5) where the relaxation is due solely to  $\delta\varepsilon$ -noise, indicating that the dominant source of energy relaxation lies elsewhere. Furthermore, the relaxation to a 300  $\Omega$  electrical environment through the measurement d.c. SQUID as estimated using a spin-boson model for the measured SQUID junction asymmetry in our device ( $<1\%$ ) is more than 100 times longer than observed<sup>36</sup>. The crossover  $f_c$  between the effective  $1/f$ - and  $f$ -type flux noises (Fig. 5) occurs near  $k_B T/\hbar$ , where  $T = 50$  mK is the approximate electronic temperature of our device<sup>26,37</sup>. Interestingly, projecting the  $1/f^\alpha$  (longitudinal) flux noise from the CPMG measurements to higher frequencies comes within a factor of two of the (transverse) flux noise at frequency  $\Delta$ ; admittedly, however,  $\alpha$  is not necessarily constant over the whole frequency spectrum.

The dynamical-decoupling protocols used in this work to carry out noise spectroscopy are the same sequences as used ubiquitously in spin-resonance applications to mitigate errors. By using the filtering property of these sequences, we have shown that it is possible to characterize directly the environmental-noise PSD, which in turn enables the design of tailored coherent control methods that target the specific environmental noise. This is particularly useful to fields such as magnetic imaging and sensing, which tend to operate in the large-error limit and are noise limited. This technique can also bring clarity to the origin of the noise itself. Although speculative, we note for further study that the PSD power law obtained experimentally by the CPMG technique, when extended to higher frequencies, falls within a factor of two of the measured transverse noise, suggesting the possibility that the microscopic mechanism responsible for low-frequency dephasing may also play a role in high-frequency relaxation. In the low-error limit, integrating optimized refocusing pulses into qubit control sequences, for example, by forming composite gates that incorporate both quantum operations and refocusing pulses, may lead to lower net error rates in systems limited by dephasing<sup>38</sup>. Despite observing levels of  $1/f$  flux and critical-current noise similar to those observed ubiquitously in superconducting qubits and SQUIDS<sup>21,39</sup>, we could mitigate this noise dynamically to increase the pure dephasing times  $T_\varphi$  beyond 0.1 ms, more than a factor of  $10^4$  longer than the intrinsic pulse length. However, despite having a remarkably long energy-relaxation time  $T_1 = 12 \mu\text{s}$ , the transverse relaxation  $T_2 \approx 2T_1$  was ultimately limited by it. Dynamical-decoupling protocols go a long way to refocusing existing levels of  $1/f$  noise and achieve long coherence times, and the main emphasis is now on identifying and mitigating the noise source(s) that cause energy relaxation.

Received 25 February 2011; accepted 4 April 2011;  
published online 8 May 2011

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## Acknowledgements

We gratefully acknowledge T. Orlando for support in all aspects of this work. We appreciate M. Biercuk, J. Clarke, L. Levitov and S. Lloyd for discussions, and P. Forn-Diaz and S. Valenzuela for comments on the manuscript. We thank P. Murphy and the LTSE team at MIT Lincoln Laboratory for technical assistance. This work was sponsored by the US Government, the Laboratory for Physical Sciences, the National Science Foundation and the Funding Program for World-Leading Innovative R&D on Science and Technology (FIRST), CREST-JST, MEXT kakenhi 'Quantum Cybernetics'. Opinions, interpretations, conclusions and recommendations are those of the author(s) and are not necessarily endorsed by the US Government.

## Author contributions

F. Yoshihara, K.H., Y.N. and J.-S.T. designed and fabricated the device. J.B. and S.G. carried out the experiments. G.F., S.G. and J.B. contributed to the software infrastructure. J.B., F. Yan, W.D.O. and Y.N. analysed the data and F. Yoshihara and D.G.C. provided feedback. J.B. and W.D.O. wrote the paper with feedback from all authors. W.D.O. supervised the project. All authors contributed to discussions during the conception, execution and interpretation of the experiments.

## Additional information

The authors declare no competing financial interests. Supplementary information accompanies this paper on [www.nature.com/naturephysics](http://www.nature.com/naturephysics). Reprints and permissions information is available online at <http://www.nature.com/reprints>. Correspondence and requests for materials should be addressed to J.B.

# Noise spectroscopy through dynamical decoupling with a superconducting flux qubit

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April 15, 2011

## Contents

|          |                                                                                |          |
|----------|--------------------------------------------------------------------------------|----------|
| <b>A</b> | <b>Measurement set-up</b>                                                      | <b>2</b> |
| <b>B</b> | <b>Description of the qubit</b>                                                | <b>3</b> |
| <b>C</b> | <b>Description of the read-out SQUID</b>                                       | <b>4</b> |
| <b>D</b> | <b>Sensitivity to noise</b>                                                    | <b>5</b> |
| <b>E</b> | <b>Decoherence. Ramsey and echo phase decays — <math>1/f</math>-flux noise</b> | <b>6</b> |
| <b>F</b> | <b>Numerical simulations of Ramsey and echo</b>                                | <b>7</b> |
| <b>G</b> | <b>Pulse calibration</b>                                                       | <b>8</b> |
| <b>H</b> | <b>Dynamical-decoupling pulse sequences</b>                                    | <b>8</b> |
| <b>I</b> | <b>Determination of the PSD from the CPMG-decay data</b>                       | <b>9</b> |
| I.1      | Decay during pulses . . . . .                                                  | 9        |
| I.2      | Relaxation . . . . .                                                           | 10       |
| I.3      | Dephasing . . . . .                                                            | 10       |



## B Description of the qubit

We fabricated our device at NEC, using the standard Dolan angle-evaporation deposition process of Al–AlO<sub>x</sub>–Al on a SiO<sub>2</sub>/Si wafer (Supplementary Fig. 1a).

The persistent-current, or flux qubit [2–5] consists of a superconducting loop with diameter  $d \sim 2 \mu\text{m}$ , interrupted by four Josephson junctions (Supplementary Fig. 1). Three of the junctions each have the Josephson energy  $E_J = 210 \text{ GHz}$ , and charging energy  $E_C = 4 \text{ GHz}$ ; the fourth is smaller by a factor  $\alpha = 0.54$ . The ratio of energy scales puts the device in the flux limit,  $E_J/E_C \approx 50$ , thus making the phases across the Josephson junctions well defined. The geometric and kinetic loop inductances are negligible compared to the Josephson inductance:

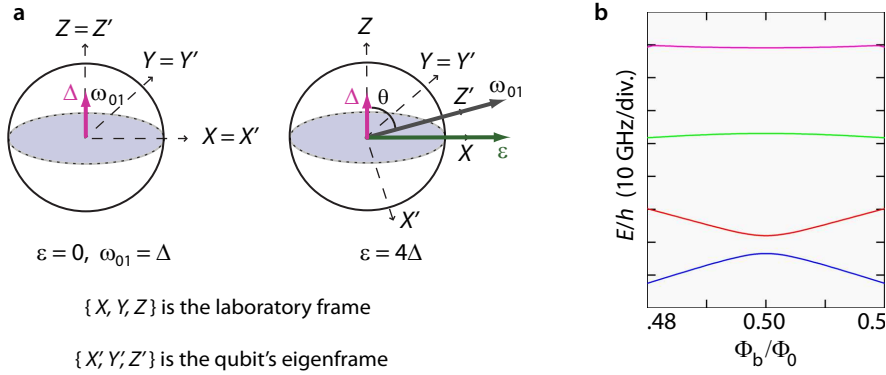
$$L_g \sim \mu_0 d \sim 2 \text{ pH}, \quad L_k = \mu_0 \lambda_L^2 l/S \sim 30 \text{ pH}, \quad L_J = \Phi_0/2\pi I_c \sim 10 \text{ nH},$$

where we used  $\lambda_L = 100 \text{ nm}$ ,  $l = 10 \mu\text{m}$ , and  $S = 20 \times 250 \text{ nm}^2$ .

When the external magnetic flux  $\Phi$  threading the qubit loop is close to half a magnetic-flux quantum,  $\Phi_0/2$ , the qubit's potential energy exhibits a double-well profile with quantized energy levels  $\pm \hbar \varepsilon/2 = \pm I_p \Phi_b$ , where  $\Phi_b = \Phi - \Phi_0/2$  is the flux bias and  $I_p = 0.18 \mu\text{A}$  the persistent current. The ground states in the left and right potential wells correspond to the diabatic states  $|L\rangle$  and  $|R\rangle$  of opposite circulating persistent currents. A tunnel coupling  $\hbar \Delta/2$  between the right- and left-well qubit states opens an energy gap,  $\hbar \Delta$ , between the ground and excited states,  $|0\rangle$  and  $|1\rangle$ , at flux degeneracy,  $\Phi_b = 0$ . The two-level Hamiltonian – analogous to a spin-1/2 particle in a magnetic field – is in the “laboratory” frame

$$\hat{\mathcal{H}} = -\frac{\hbar}{2} [(\varepsilon + \delta\varepsilon)\hat{\sigma}_x + (\Delta + \delta\Delta)\hat{\sigma}_z], \quad (1)$$

where  $\hbar \omega_{01} = \hbar \sqrt{\varepsilon^2 + \Delta^2}$  is the energy-level splitting;  $\hat{\sigma}_j$  are the Pauli matrices; and  $\delta\varepsilon$  and  $\delta\Delta$  are the noise fluctuations. The quantization axis makes an angle  $\theta = \arctan(\varepsilon/\Delta)$  with  $\hat{\sigma}_z$  (and an angle  $\pi - \theta$  with the persistent-current eigenstates), so that the qubit is first-order insensitive to flux noise when biased at the degeneracy point  $\varepsilon = 0$ . (We write the Hamiltonian in this way rather than with swapped  $x$  and  $z$  indices as in several previous papers, so that our pulses will be along  $X$  and  $Y$  in agreement with the habitual NMR language.) It can,



Supplementary Figure 2: **Two-level Hamiltonian in different eigenbases, and simulated energy spectrum.**

**a.** The Hamiltonian (1) in the “laboratory” frame  $\{X, Y, Z\}$  has the  $\varepsilon$  field (flux-bias) along  $X$ , perpendicular to the plane of the qubit loop, and the fixed tunnel coupling  $\Delta$  along  $Z$ . The Hamiltonian (2) in the qubit's eigenframe  $\{X', Y', Z'\}$  makes an angle  $\theta$  with  $Z$ . The figure shows both frames under two bias conditions ( $\varepsilon = 0$  [ $\theta = 0^\circ$ ] and  $\varepsilon = 4\Delta$ ), with the laboratory frame fixed in space. The two frames coincide when  $\varepsilon = 0$ . **b.** Simulated spectrum.

alternatively, be written in the qubit's eigenbasis,

$$\hat{\mathcal{H}} = -\frac{1}{2}\hbar(\omega_{01}\hat{\sigma}_{z'} + \delta\omega_{z'}\hat{\sigma}_{z'} + \delta\omega_{\perp'}\hat{\sigma}_{\perp'}), \quad (2)$$

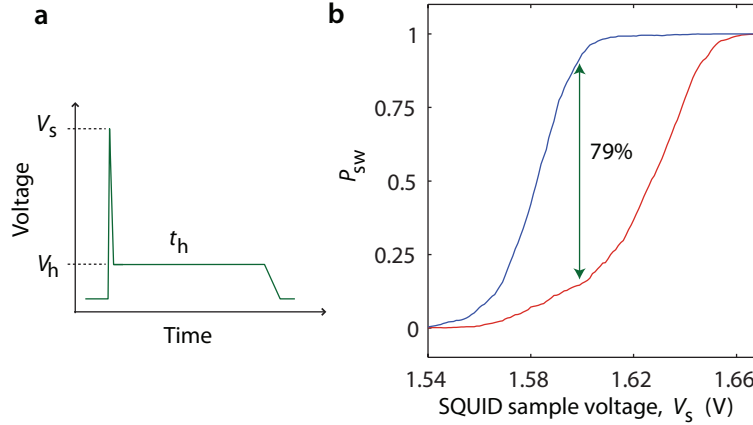
see Supplementary Fig. 2a. Here  $\hat{\sigma}_{\perp'}$  denotes that the transverse spin component can include both  $\hat{\sigma}_{x'}$  and  $\hat{\sigma}_{y'}$ . The longitudinal and transverse noises,  $\delta\omega_{z'}$  and  $\delta\omega_{\perp'}$ , are further described below.

A coil with mutual coupling  $M_{\text{dc}} = 0.6$  pH to the qubit loop provides the dc flux bias. The qubit loop is also coupled to an on-chip microwave antenna (aluminium loop launched from a coplanar waveguide towards ground) with designed mutual inductance  $M = 0.02$  pH ( $R = 50\ \Omega$  termination). This value corresponds very closely to the one inferred from a measurement of the Rabi frequency for a known applied microwave power. An amplitude of 50 mV at the generator and a total attenuation of about 56 dB corresponds to about  $-69$  dBm at the device, or an antenna current of  $I_{\mu\text{w}} \approx 1.6\ \mu\text{A}$ . The corresponding Rabi frequency is  $\Omega_{\text{R}}/2\pi = 16$  MHz, which is generated by a flux  $\Phi_{\text{R}} = \hbar\Omega_{\text{R}}/2I_{\text{P}} = 14\ \mu\Phi_0$ . The mutual inductance follows:  $M = \Phi_{\text{R}}/I_{\mu\text{w}} \approx 0.02$  pH.

As shown in Fig. 1 of the main paper, we measured remarkably long coherence times  $T_1$ ,  $T_2$  and  $T_2^*$  in this device. In nominally identical devices, we typically measure similar values for the strength of the low-frequency,  $1/f$ -type noise as in this device (section E below). In contrast, the energy-relaxation times are not as reproducible between devices; other samples from the current batch had shorter  $T_1$ .

## C Description of the read-out SQUID

Our hysteretic dc-SQUID [6], see Supplementary Fig. 1, has critical current  $I_{\text{c}} = 4.5\ \mu\text{A}$ ; normal resistance  $R_{\text{N}} = 0.25\ \text{k}\Omega$ ; mutual qubit-SQUID inductance  $M_{\text{Q-S}} = 21$  pH; on-chip capacitors  $C \sim 10$  pF, inductors  $L \sim 0.1$  nH, and bias resistors  $R_{\text{I}} = 0.2\ \text{k}\Omega$  and  $R_{\text{V}} = 1\ \text{k}\Omega$ ; and further cold and room-temperature resistors and filters. The shunt capacitors bring the plasma frequency down to  $\omega_{\text{p}}/2\pi = 2.1$  GHz.



Supplementary Figure 3: **Read-out pulse and visibility.** **a**, Read-out pulse (voltage across a 1-k $\Omega$  bias resistance). The optimized pulse shape is programmed to have a 4-ns sample pulse, including 1-ns rise, 1-ns sample, and 2-ns fall times, followed by a 3- $\mu\text{s}$  hold plateau at 17.5% of the sample voltage. **b**, Read-out visibility at  $\Phi_{\text{b}} = 0$ . Scans of the SQUID's switching probability,  $P_{\text{sw}}$ , vs. the height of the sampling pulse. We obtain 79 % read-out visibility between the qubit's ground and excited states. Relaxation during read leads to an imperfect determination of the excited state (17 % dark counts at the optimal  $V_{\text{s}}$ ).

The qubit's magnetization, resulting from the persistent currents, modulates the SQUID's switching current. For qubit read out, we apply a sample-and-hold current pulse to the SQUID (Supplementary Fig. 3a) and, after room-temperature amplification, use a threshold detector to register the presence (absence) of a voltage corresponding to the qubit being (not being) in state  $|L\rangle$ . We determine the switching probability,  $P_{\text{sw}}$ , statistically by repeating this measurement several thousand times.

The pulse is produced by a digital arbitrary-waveform generator (Tektronix AWG5014) with 250 MHz analog output bandwidth. Due to enhanced relaxation when a current is flowing through the SQUID, a rapid sample pulse is important for good read-out visibility. A hold current enables the room-temperature electronics to register the voltage pulse before retrapping occurs, but a low and short hold pulse limits the quasiparticle generation, which is also beneficial for the visibility. We obtain an optimal read-out visibility of 79% (Supplementary Fig. 3b).

In addition to biasing the SQUID, the read-out pulse couples flux into the qubit with the effect of shifting the states adiabatically,  $|0\rangle \rightarrow |L\rangle$  and  $|1\rangle \rightarrow |R\rangle$ , before the measurement takes place, so that the states can be distinguished in the persistent-current basis.

The optimal SQUID dc-bias current  $I_b$  with respect to noise coupled into the qubit was very close to  $I_b = 0$ , indicating highly symmetric SQUID junctions. In this set of experiments, we therefore did not apply any dc bias to the SQUID while manipulating the qubit, *cf.* Ref. [1].

## D Sensitivity to noise

Following the approach of Ithier *et al.* [7], we will evaluate the noise terms in the qubit's Hamiltonian,

$$\hat{\mathcal{H}} = -\frac{1}{2}\hbar(\omega_{01}\hat{\sigma}_{z'} + \delta\omega_{z'}\hat{\sigma}_{z'} + \delta\omega_{\perp'}\hat{\sigma}_{\perp'}) \quad [\text{equation (2)}].$$

We consider both flux and critical-current noise, and express the noises as sensitivity derivatives that translate the noise fluctuations to a change in the Hamiltonian, *i.e.*  $\delta\lambda \rightarrow \delta\omega$ . At  $\Phi_b = 0$ , the first-order noise  $\delta\Delta$  is much larger than second-order  $\delta\varepsilon$  (flux) noise, as confirmed through simulation. It is therefore sufficient to expand to first order to explain our Ramsey and echo data,

$$\delta\omega_{z'} = \frac{\partial\omega_{01}}{\partial\lambda}\delta\lambda + \dots \quad \text{and} \quad \delta\omega_{\perp'} = \frac{\partial\omega_{\perp'}}{\partial\lambda}\delta\lambda + \dots \quad (3)$$

Energy relaxation is related to noise that is transverse to the qubit's quantization axis,  $\delta\omega_{\perp'}(\lambda)$ , at the frequency of the level splitting  $\omega_{01}$ .

Pure dephasing, on the other hand, is related to low-frequency, longitudinal fluctuations of the qubit's energy-level splitting. To evaluate the longitudinal first-order term in equation (3) we use the chain rule,

$$\frac{\partial\omega_{01}}{\partial\lambda} = \frac{\partial\omega_{01}}{\partial\varepsilon} \frac{\partial\varepsilon}{\partial\lambda} + \frac{\partial\omega_{01}}{\partial\Delta} \frac{\partial\Delta}{\partial\lambda}. \quad (4)$$

By geometry,

$$\frac{\partial\omega_{01}}{\partial\varepsilon} = \frac{\varepsilon}{\omega_{01}} \quad \text{and} \quad \frac{\partial\omega_{01}}{\partial\Delta} = \frac{\Delta}{\omega_{01}}. \quad (5)$$

From spectroscopy measurements we infer the  $\varepsilon$  sensitivity to flux noise,  $\lambda = \Phi$ ,

$$\kappa_{\varepsilon,\Phi} \equiv \kappa_{\varepsilon} = \frac{\partial\varepsilon}{\partial\Phi} = 2\pi \times (1.1 \text{ GHz/m}\Phi_0), \quad (6)$$

while  $\Delta$  is insensitive to  $\Phi$  noise.

Since  $\hbar\varepsilon = 2I_p\Phi_b$  and  $I_p = I_c\sqrt{1 - 1/(2\alpha)^2}$ ,  $\varepsilon$  is sensitive also to critical-current noise,  $\lambda = I_c$ ,

$$\kappa_{\varepsilon,I_c} = \frac{\partial\varepsilon}{\partial I_c} = \frac{\partial\varepsilon}{\partial I_c/I_c} = \frac{\varepsilon}{I_c} I_c = \varepsilon. \quad (7)$$

For the  $\Delta$  sensitivity to fluctuations in  $i_c = \delta I_c / I_c$ , a numerical simulation gives  $\partial\Delta/\partial I_c = 2\pi \times (8 \cdot 10^{15} \text{ Hz/A})$ , and with  $I_c = 0.4 \mu\text{A}$  we obtain

$$\kappa_{\Delta, i_c} \equiv \kappa_{\Delta} = \frac{\partial\Delta}{\partial i_c} = 2\pi \times (3.2 \text{ GHz}). \quad (8)$$

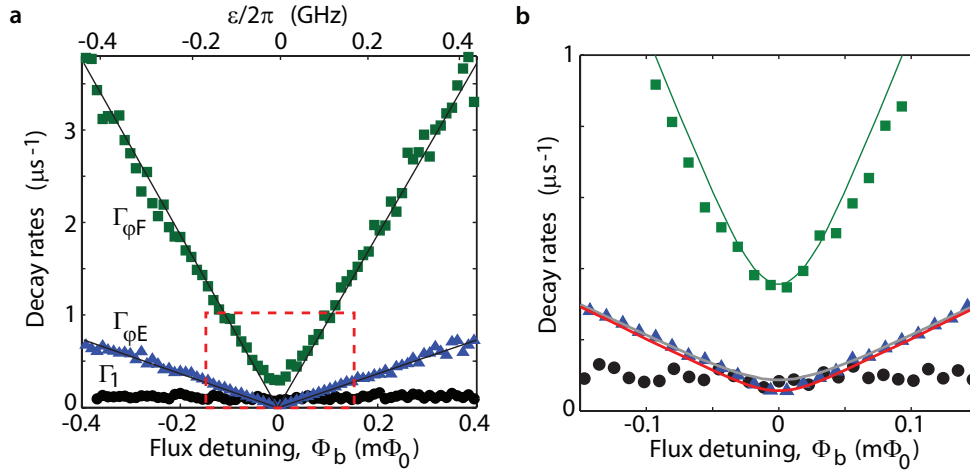
Taken together, the first-order fluctuations are therefore

$$\frac{\partial\omega_{01}}{\partial\lambda} \delta\lambda = \frac{\varepsilon}{\omega_{01}} \kappa_{\varepsilon, \Phi} \delta\Phi + \left( \frac{\varepsilon}{\omega_{01}} \kappa_{\varepsilon, i_c} + \frac{\Delta}{\omega_{01}} \kappa_{\Delta, i_c} \right) \delta i_c \quad (9)$$

Flux noise dominates, except very near  $\varepsilon = 0$ , where the  $\delta\Delta$  noise is needed to account for the observed decay (the first term is first order in  $\varepsilon$  and the second term is negligible because it is of order  $\varepsilon^2$ ).

## E Decoherence. Ramsey and echo phase decays — $1/f$ -flux noise

In the Bloch–Redfield formalism, valid for weakly coupled, short-correlated noise, the dynamics of two-level systems is described by the longitudinal and transverse relaxation rates,  $\Gamma_1 = 1/T_1$  and  $\Gamma_2 = 1/T_2$ , respectively [7, 8]. We assume that the qubit is coupled to many fluctuators, which, in concert and independent of their individual statistics, will yield a Gaussian noise distribution due to the central limit theorem. In systems where the noise is regular at the frequency of the qubit’s energy-level splitting,  $\omega_{01}$ , relaxation and dephasing factorize. The longitudinal relaxation is exponential due to the many uncorrelated contributions of transversally coupled noise at  $\omega_{01}$ . The pure-dephasing rate,  $\Gamma_{\varphi}$ , associated with low-frequency, quasi-static, longitudinally coupled noise (inhomogeneous broadening), combines with  $\Gamma_1$  to give the rate  $\Gamma_2 = \Gamma_1/2 + \Gamma_{\varphi}$ . However, the exact form of



Supplementary Figure 4: **Decay rates vs. flux detuning.** **a**, Energy-relaxation rate,  $\Gamma_1$ , and Gaussian phase-decay rates for the echo,  $\Gamma_{\varphi, E}$ , and free induction (Ramsey),  $\Gamma_{\varphi, F}$ , after subtracting the exponential  $\Gamma_1$  decay. Straight, black lines are fits to  $\Gamma_{\varphi, E/F}(\Phi_b)$  for flux noise only, see text and equation (10). **b**, Blow up of the data in the dashed-box region in **a**, along with simulated Ramsey- and echo- decay rates including both  $\delta\varepsilon$  and  $\delta\Delta$  noises. For the echo, the grey line is the rate obtained for a  $1/f$  noise in  $\delta\Delta$  with the 1-MHz high-frequency cut off given by the experimental protocol. The red line is the rate obtained with the lower high-frequency cut off:  $\omega_{\text{high}}^{\lambda}/2\pi < 0.1 \text{ MHz}$ . The exact rate depends on the detailed shape of the noise cut off, which is unknown.

the time dependence of the dephasing component is determined by the noise-PSD. (That is, when the dephasing is non-exponential, the inverse time constant can strictly no longer be interpreted as a simple rate.)

At  $\Phi_b = 0$ , the echo decay is nearly  $T_1$ -limited in our device and, therefore, practically indistinguishable from an exponential. Relaxation contributes by  $\Gamma_1/2 = 1/2 T_1 = 42 \times 10^3/\text{s}$  or 6.6 kHz to the low-power spectroscopic line width  $\Delta f_{(\text{FWHM})} = 180 \text{ kHz}$ . This exceeds the expected  $\Delta f_{(\text{FWHM})} = 1/\pi T_2^*$  by only 50 kHz, indicating little power broadening, given the free-induction decay rate,  $\Gamma_F = 1/T_2^*$ , with the measured  $T_2^* = 2.5 \mu\text{s}$ .

Biased away from  $\Phi_b$ , we find a Gaussian spin-echo phase decay,  $\exp[-(\Gamma_{\varphi,E} t)^2]$ , consistent with a  $1/f$  flux-noise model [7] (singular PSD at  $\omega = 0$ ). As the Ramsey fringe decays considerably faster, it is hard to distinguish between purely exponential and Gaussian decays. The bias dependencies for echo and Ramsey decays are

$$\Gamma_{\varphi,E/F}(\Phi_b) = (A_\Phi \eta_{E/F})^{1/2} \left| \frac{\partial \omega_{01}}{\partial \Phi} \right|, \quad (10)$$

where the numerical factors  $\eta$  differ due to the different echo and Ramsey filtering functions:  $\eta_{\varphi,F} = \ln(1/\omega_{\text{low}} t)$  and  $\eta_{\varphi,E} = \ln 2$ . Their ratio,  $\Gamma_{\varphi,F}(\Phi_b)/\Gamma_{\varphi,E}(\Phi_b) \approx 4.5$ , is in accordance with our data, and the magnitude,

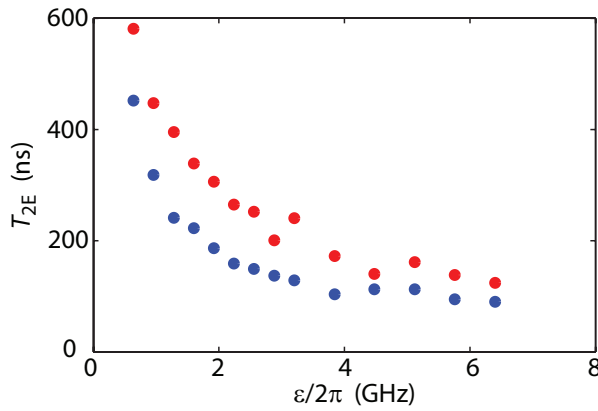
$$A_\Phi = (1.7 \mu\Phi_0)^2, \quad (11)$$

of the noise  $S(\omega) = A_\Phi/|\omega|$  agrees with previous results [1], see Supplementary Fig. 4.

The range in flux detuning covered in Supplementary Fig. 4 corresponds to a change in qubit frequency of 15 MHz. However, as shown in Supplementary Fig. 5, we can improve  $T_2$  by applying  $\pi$  pulses over a much wider range. In fact, the derivative in equation (10) reaches an asymptotic value as  $\Phi_b$  grows, and we expect the dephasing value to saturate at  $T_{2E} = 100 \text{ ns}$ . Although the Ramsey  $T_2^*$  is near zero this far detuned, the echo protocol works well, since in our qubit  $T_{2E}$  is much longer than the time  $\tau_\pi$  of one pulse.

## F Numerical simulations of Ramsey and echo

The  $\Phi_b$  dependencies of  $T_2^*$  and  $T_{2,E}$  are reproduced in numerical simulations. We simulated the Ramsey fringe by numerically solving the Bloch equations, including the measured  $T_1$  decay and linearly coupled quasi-static noises, averaging over many realizations. Each run had stochastic, normally distributed  $\delta\varepsilon$  and  $\delta\Delta$  deviations from



Supplementary Figure 5: **Echoes at large flux detuning**, Blue and red dots denote echoes with one  $\pi$  pulse and two  $\pi$  pulses, respectively (CPMG sequence, see section H below). The ratio  $T_{2E}^{(2\pi)}/T_{2E}^{(1\pi)}$  is nearly constant around 1.5 over this range.

their average values: The  $1/f$  flux noise,  $S(\omega) = A_\Phi/|\omega|$ , gives  $\sigma_\varepsilon/2\pi = 10$  MHz when integrated from 1 Hz to 1 MHz; noise in  $\Delta$  dominates near  $\Phi_b = 0$ , and we get agreement for  $\sigma_\Delta/2\pi = 0.06$  MHz, uncorrelated with the flux noise and obtained from  $S_\Delta(\omega) = A_\Delta/|\omega|$  with  $A_\Delta = (4 \cdot 10^{-6})^2$ .

In the echo simulation we additionally took into account dynamic noises from  $10^4$  to  $10^{10}$  Hz. We obtained the noisy time series for  $\varepsilon$  and  $\Delta$  by inverse-Fourier transforming the amplitudes of the  $1/f$  noises with random phases for each Fourier component, and then evaluated the Schrödinger evolution operator in discrete time.

Near  $\Phi_b = 0$ , the experimental data shows a greater echo-improvement in  $T_2$  than the simulation would suggest for these parameters (see Supplementary Fig. 4b). This discrepancy would be explained by a lower high-frequency  $\Delta$ -noise cut off, with faster decay than  $1/f$  above  $\omega_{\text{high}}^\Delta/2\pi \sim 0.1$  MHz.

Numerical evaluations of the coherence integral, equation (14) below, agree with our simulations.

## G Pulse calibration

We calibrate the rotations to  $< 1\%$  accuracy by applying a tune-up sequence of pulses, akin to methods used in NMR [9]. A rigorous measurement of gate errors should be done with randomized benchmarking, for example.

## H Dynamical-decoupling pulse sequences

Collin *et al.* [10] and Ithier *et al.* [7] have employed some NMR methods beyond the Hahn spin echo to manipulate a superconducting qubit. In this work, we apply multi-pulse, dynamical-decoupling pulse sequences to facilitate spectroscopy of the environmental noise and to significantly enhance the coherence times.

During the Carr-Purcell-Meiboom-Gill (CPMG) multi-echo sequence [11, 12], defined as

$$X_{\pi/2} - \left( \frac{\tau}{2N} - Y_\pi - \frac{\tau}{2N} \right)_N - X_{\pi/2} \quad (12)$$

and illustrated by the rotations in the Bloch sphere (Supplementary Fig. 6), the transverse component of the Bloch vector is refocused along the axis of the refocusing pulses (here  $Y$ ), whereas the perpendicular component ( $X$ ) is randomized. CPMG is a development on the Carr-Purcell (CP) sequence [11], which has identical pulse positions, but where all rotations are along the same axis. The third pulse sequence that we investigated, the Uhrig dynamical decoupling (UDD) sequence [13, 14], has normalized pulse positions defined by

$$\delta_j = \sin^2 \left( \frac{\pi j}{2N+2} \right). \quad (13)$$

The dephasing under a certain sequence is described by the *coherence integral*,

$$\chi_N(\tau) = \left( \frac{\partial \omega_{01}}{\partial \lambda} \right)^2 \tau^2 \int_0^\infty d\omega S(\omega) g_N(\omega, \tau). \quad (14)$$

Here the dimensionless filter function  $g_N(\omega, \tau)$  depends on the number and distribution of the  $\pi$  pulses [13–17],

$$g_N(\omega, \tau) = \frac{|y_N(\omega, \tau)|^2}{(\omega\tau)^2}, \quad (15)$$

$$y_N(\omega, \tau) = \left| 1 + (-1)^{1+N} \exp(i\omega\tau) + 2 \sum_{j=1}^N (-1)^j \exp(i\omega\delta_j\tau) \cos(\omega\tau_\pi/2) \right|, \quad (16)$$

where  $\delta_j \in [0, 1]$  is the normalized position of the centre of the  $j$ th  $\pi$  pulse between the two  $\pi/2$  pulses. The  $\tau_\pi$ -dependent factor assumes square pulses, but is a reasonable approximation in our case. Alternatively to equation (15), one can define a function  $F_N(\omega, \tau) = |y_N(\omega, \tau)|^2$  that filters the phase noise,  $S(\omega)/\omega^2$ , as in, *e.g.*, Refs. [13–17].

The Ramsey free induction ( $N = 0$ ) and the Hahn spin echo ( $N = 1$ ) have weighting functions  $g_0(\omega, \tau) = \text{sinc}^2(\omega\tau/2)$  and  $g_1(\omega, \tau) = \text{sinc}^2(\omega\tau/4) \sin^2(\omega\tau/4)$ , respectively. Note that  $g_1(0) = 0$ , indicating a suppression of the low-frequency part of the noise for the Hahn echo.

The CPMG sequence is inherently robust to field inhomogeneities when the following criteria are fulfilled [18]: (i) The effective rotation axis is oriented in the transverse ( $XY$ ) plane (true when the driving frequency is resonant with the level splitting); (ii) the magnetization (Bloch vector) is initially aligned with the rotation axis ( $Y$ ); and (iii) the rotation angle is  $\pi$ . Errors that occur due to deviations from (i–iii) can be quantified by computing the propagator over one cycle of the sequence. For an initial state  $Y$ , one finds that  $Y_\pi$  errors appear only to fourth order (CPMG), whereas  $X_\pi$  errors accumulate to second order (CP).

## I Determination of the PSD from the CPMG-decay data

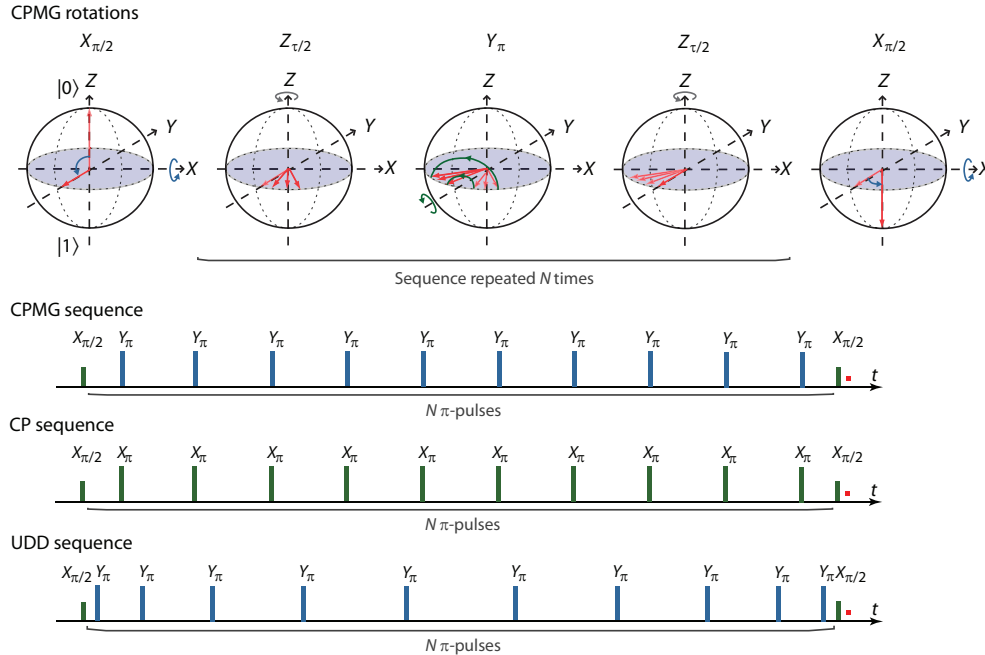
The qubit is subjected to different decay mechanisms during the CPMG pulse sequence. We identify three decay rates, and fit the qubit's population to the function

$$P_{\text{sw}}(\tau) = P_0 + a \exp(-\Gamma_1 \tau/2) \exp(-\Gamma_p(\tau_p)) \exp(-\chi_N(\tau)), \quad (17)$$

keeping the constants  $a$  and  $P_0$  fixed for all  $N$ .

### I.1 Decay during pulses

There is pulse-induced decoherence during the total pulse time  $\tau_p = N\tau_\pi + 2\tau_{3\pi/2}$ , in addition to  $T_1$  decay. (For experimental reasons, instead of  $\pi/2$  pulses we often use the functionally equivalent  $3\pi/2$  pulses.) We do not



Supplementary Figure 6: **Dynamical-decoupling pulse sequences.** Rotations of the Bloch vector during the CPMG sequence (pulses along  $X$  and  $Y$  in the laboratory frame.) Timing of the CPMG, CP, and UDD pulse sequences for  $N = 10$ .

need to assume that this decay has any particular form, *e.g.* exponential, as it will be divided out of the calculation. However, a simple exponential decay with  $\Gamma_p = 1/T_p$  where  $T_p = 1.75 \mu\text{s}$  gives a good overall fit independent of  $N$ . Although much higher than the Rabi-decay rates described in the manuscript, this rate is reasonable given that we observe an increased Rabi-decay time at high driving amplitudes (data not shown).

## I.2 Relaxation

For the duration of the sequence, there is energy relaxation with the constant rate  $\Gamma_1/2 = 1/2 T_1$ , where  $T_1 = 12 \mu\text{s}$ . We put the  $T_1$  decay during the pulses into the function  $\Gamma_p$  and therefore  $\Gamma_1/2$  in equation (17) is acting only during the free-evolution time  $\tau$ .

## I.3 Dephasing

The function  $\chi_N(\tau)$  depends on the dephasing rate during the total free-evolution time  $\tau$  under an  $N$ -pulse CPMG sequence, which provides us with a means of figuring the noise-PSD,  $S(\omega)$ . Only the PSD appears in the decay function, because the noise statistics were taken to be Gaussian, and all information is included in the second-order correlation function.

Dividing equation (17) with itself for two free-evolution times,  $\tau_1$  and  $\tau_2$ , we get

$$q(\tau_1, \tau_2) = \frac{P_{\text{sw}}(\tau_1) - P_0}{P_{\text{sw}}(\tau_2) - P_0} = \exp\left(-\Gamma_1[\tau_1 - \tau_2]\right) \exp\left(-[\chi_N(\tau_1) - \chi_N(\tau_2)]\right). \quad (18)$$

For a pulse separation  $\tau' = \tau/N$  (and large  $N$ ), the filter (15) is peaked near

$$\omega'(\tau') = \frac{2\pi}{2\tau'}. \quad (19)$$

It is narrow enough, that we can treat  $g_N(\omega'(\tau'), \tau')$  as a delta function peaked at  $\omega'$ , and regard the noise in the qubit's transition frequency as constant within its bandwidth,  $\Delta\omega$ , as illustrated in Supplementary Fig. 7a. This provides us with a means of characterizing the noise at a frequency  $\omega$  by varying  $\tau$ . We can rewrite the integral in equation (14) as  $\int_0^\infty d\omega g_N(\omega'(\tau'), \tau') \rightarrow \Delta\omega \int_0^\infty d\omega \delta(\omega - \omega')$ , and do the approximation

$$\chi_N(\tau) \approx \left(\frac{\partial\omega_{01}}{\partial\lambda}\right)^2 \tau^2 S(\omega'(\tau')) g_N(\omega'(\tau'), \tau') \Delta\omega. \quad (20)$$

We know  $\Delta\omega(\tau)$ , and  $g_N(\omega'(\tau'), \tau')$  from a numerical calculation for each  $N$ .

We can use equation (20) to evaluate the second factor on the right-hand side of equation (18),

$$\tilde{q}(\tau_1, \tau_2) = \exp\left(-\left(\frac{\partial\omega_{01}}{\partial\lambda}\right)^2 \left[\tau_1^2 S(\omega'(\tau_1)) g_{N,1}(\omega'(\tau_2), \tau_2) \Delta\omega_1 - \tau_2^2 S(\omega'(\tau_2)) g_N(\omega'(\tau_2), \tau_2) \Delta\omega_2\right]\right). \quad (21)$$

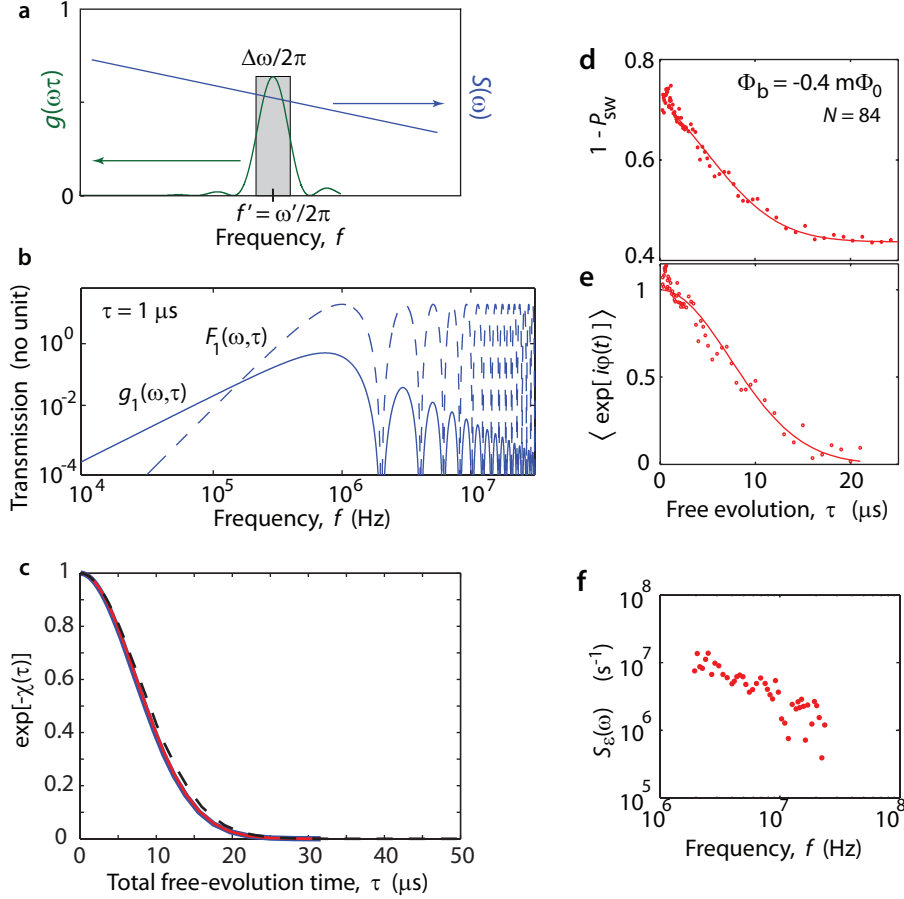
This gives us the difference between the noises at  $\omega'(\tau_1)$  and  $\omega'(\tau_2)$ , so that we can find one if we know the other. To get the absolute noise level, we have to find  $S(\omega)$  for some value of  $\omega' = \omega'(\tau')$  as a starting point. To do this, for the first point, we fix the rates  $\Gamma_p$ ,  $\Gamma_1$  and fit the dephasing using a Gaussian decay function  $\chi_N(\tau) = (\Gamma_\varphi \tau)^2$ . From that point onwards, we recursively find the remainder of the PSD using equation (20) without presuming a function form for the decay or the PSD.

Following this procedure, we obtain a number of points representing the noise PSD from 0.2 to 20 MHz. The points within the range 0.2–2 MHz lie tight together and are the least susceptible to error, which is why we fit this data to a straight line, rendering the reported  $1/f^\alpha$  dependence:  $A'_\epsilon = (0.8 \mu\Phi_0)^2$  and  $\alpha = 0.9$ . Integrating the

noise PSD from these parameters yields  $\sigma_\varepsilon = 8$  MHz (compared to 10 MHz) for the quasi-static noise, slightly changing the slope of the calculated  $\Gamma_{\varphi, F}$  in Supplementary Fig. 4, which presumed  $\alpha = 1$ . However, the slope of the PSD is not necessarily constant over many orders of magnitude, which could explain the slight deviation for the Ramsey, being sensitive to low-frequency noise.

Supplementary Figure 7c shows a numerical evaluation of the coherence integral (14), assuming a noise-PSD

$$S(\omega) = A/\omega. \quad (22)$$



Supplementary Figure 7: **Figuring the noise-PSD.** **a**, The filter function is narrow, and the noise PSD is smooth enough, that we can obtain the noise PSD at the filter's centre frequency by evaluating the coherence integral  $\chi_N(\tau)$ , equation (20). **b**, Comparison, for  $N = 1$ , of the band-pass envelope filter function  $g_N(\omega, \tau)$  used in this work to facilitate the determination of the noise-PSD (*cf.* **a**), and the high-pass envelope filter function  $F_N(\omega, \tau) = g_N(\omega, \tau) \times (\omega\tau)^2$  used in several previous works, see text. **c–f**, Fitted phase decay,  $\exp[-\chi_N(\tau)]$ , under an  $N = 84$   $\pi$ -pulse CPMG sequence with  $\tau_\pi = 6.8$  ns, at the flux detuning  $\varepsilon/2\pi = -430$  MHz ( $\Phi_0 = -0.4$  m $\Phi_0$ ), *cf.* Fig. 3b in the main text. **c**, **Blue** is the function  $\exp(-(\Gamma_\varphi^{(N)}\tau)^2)$  with a fitted  $\Gamma_\varphi^{(N)} = (8.5 \mu\text{s})^{-1}$ . **Red** is the integral (14) with  $S(\omega) = A/\omega$ . Note the excellent agreement between the blue and red lines. **Dashed** is the expression (20) with the same parameters as for the red line. **d**, Population decay under an 84-pulse CPMG sequence. Gaussian fit with fixed exponential decay contributions  $T_1 = 12 \mu\text{s}$ ,  $T_p = 1.75 \mu\text{s}$ . **e**, Phase decay of the signal in **d** after dividing out  $T_1$  and  $T_p$ . **f**, Noise-PSD calculated from the data in **e**.

The calculation shows, that this method underestimates the decay by about 5 %, since it disregards the part of the noise that falls within the harmonics of  $g_N$ , the first of which occurs at  $3\omega'$  and which is about 10 dB lower than the main lobe at  $\omega'$ .

## J Decoherence during driven evolution

We obtain information about the flux noise in the 2–20 MHz frequency range also by analyzing the decoherence during driven evolution.

Driven by a near-resonant, transverse field  $\hat{\mathcal{H}}_1 = -\hbar \Omega(t) \cos(\omega t) \hat{\sigma}_x$ , the static part of the Hamiltonian in the rotating frame is

$$\hat{\mathcal{H}} = -\frac{1}{2}\hbar (\Delta\omega \hat{\sigma}_z + \Omega \hat{\sigma}_x), \quad (23)$$

where  $\Delta\omega = \omega - \omega_{01}$ . As a result, the qubit oscillates at the Rabi frequency

$$\Omega_R = \sqrt{\Omega^2 + (\Delta\omega)^2} \approx \Omega + (\Delta\omega)^2/2\Omega \quad (24)$$

around the effective field, which makes an angle  $\eta = \arctan(\Delta\omega/\Omega)$  with the qubit's  $\hat{\sigma}_z$  axis.

The decay of the Rabi oscillations has contributions from noise at the frequencies  $\omega_{01}$  and  $\Omega$  [7], and from the low-frequency, quasi-static flux noise with variance  $\sigma_\varepsilon^2$ .

### J.1 Static noise

We first describe the decay due to quasi-static noise. Flux deviations from the bias point  $\varepsilon$  lead to a frequency detuning  $\Delta\omega = (\partial\omega_{01}/\partial\varepsilon)\Delta\varepsilon = (\varepsilon/\omega_{01})\Delta\varepsilon$ , normally distributed with a standard deviation  $\sigma_\varepsilon$ ,

$$N(\Delta\varepsilon) = \sigma_\varepsilon^{-1/2} \exp\left(-(\Delta\varepsilon)^2/2\sigma_\varepsilon^2\right). \quad (25)$$

Integrating over  $N(\Delta\varepsilon)$ , the Rabi oscillations decay as

$$\zeta(t) = \int d(\Delta\varepsilon) N(\Delta\varepsilon) \cos(\Omega_R t + \phi). \quad (26)$$

Under the approximation  $\Delta\omega \ll \Omega$ , the Fresnel-type integral (equation 26) becomes [7]

$$\zeta(t) = (1 + (ut)^2)^{-1/4} \cos\left(\Omega t + \frac{1}{2} \arctan(ut)\right), \quad (27)$$

where

$$u = \left(\frac{\partial\omega_{01}}{\partial\varepsilon}\right)^2 \frac{\sigma_\varepsilon^2}{\Omega} = \left(\frac{\varepsilon}{\omega_{01}}\right)^2 \frac{\sigma_\varepsilon^2}{\Omega}. \quad (28)$$

### J.2 Noise at the Rabi frequency

We now turn to the exponential decay of the oscillation envelope ( $\Gamma_2$  in the rotating frame),

$$\Gamma_R = \frac{3}{4}\Gamma_1 \cos^2 \theta + \frac{1}{2}\Gamma_\Omega, \quad (29)$$

where the component that depends on the noise at the Rabi frequency  $\Omega_R$  has two contributions which we denote  $\Gamma_\Omega^{(z)}$  and  $\Gamma_\Omega^{(\perp)}$ :

$$\Gamma_\Omega = \pi S_\varepsilon(\Omega_R) \sin^2 \theta + \pi S_\Delta(\Omega_R) \cos^2 \theta \equiv \Gamma_\Omega^{(z)} \sin^2 \theta + \Gamma_\Omega^{(\perp)} \cos^2 \theta. \quad (30)$$

We can approximate  $\cos^2 \theta \approx 1$ , for  $\varepsilon \ll \Delta$ , and write equation (29) as

$$\Gamma_R = \frac{3}{4}\Gamma_1 + \frac{1}{2}\Gamma_\Omega^{(\perp)} + \left(\frac{\varepsilon}{\omega_{01}}\right)^2 \frac{1}{2}\Gamma_\Omega^{(z)}. \quad (31)$$

### J.3 Fitting procedure

In order to determine  $S_\lambda(\Omega_R)$ , we measured the Rabi oscillations vs.  $\Phi_b$ , with  $\Omega_R$  fixed for each set of data. The combined decay envelope from equations (27, 31) becomes  $(1 + (ut)^2)^{-1/4} \exp(-\Gamma_R t)$ . We observe that, at  $\varepsilon = 0$ , the model predicts an exponential decay with the rate  $\frac{3}{4}\Gamma_1 + \frac{1}{2}\Gamma_\Omega^{(\perp)}$ . For a given  $\Omega_R$ , we find this rate by fitting, and then keep it fixed while fitting the Rabi envelopes vs.  $\varepsilon$ . The rate  $\Gamma_\Omega^{(\perp)}$  was too small to distinguish accurately from  $\Gamma_1$ , consistent with its correspondingly small quasi-static noise  $\sigma_{\lambda=\Delta}^2$  (Table 1 in the main text). At low amplitude ( $\Omega_R/2\pi \sim 2$  MHz),  $\Gamma_R$  nearly reaches its limit  $(3/4)\Gamma_1$ . Dividing out the known quasi-static contribution, and then fitting the envelope to the parabolic equation (31), we obtain the flux-noise dependent  $\Gamma_\Omega^{(z)}$ , from which we can calculate  $S_\varepsilon(\Omega)$ , see Fig. 4 in the main text.

## K Energy relaxation; high-frequency spectroscopy

Supplementary Figure 8 shows the variation over repeated measurements of  $T_1$  relaxation in this device.

To explain the high-frequency noise, leading to  $T_1$  relaxation, we model the  $\varepsilon$  and  $\Delta$  channels of the environment as ohmic resistors. The microwave antenna's designed mutual inductance to the qubit is  $M = 0.02$  pH with  $R = 50 \Omega$  termination. The Johnson–Nyquist (thermal and quantum) noise of an environment at  $T = 50$  mK (a reasonable effective electronic temperature on the chip) is<sup>1</sup>

$$S_\varepsilon^{\text{JN}}(\omega) = \frac{1}{2\pi} \left( \frac{\partial \varepsilon}{\partial \Phi} \right)^2 M^2 \frac{2\hbar\omega/R}{1 - e^{-\hbar\omega/k_B T}}, \quad (32)$$

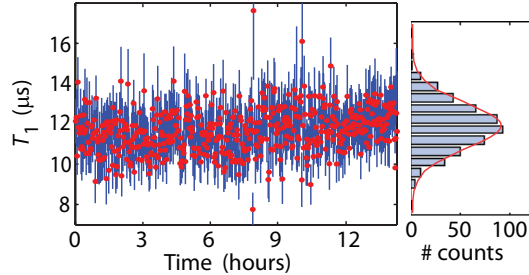
which simplifies to

$$S_\varepsilon^{\text{JN}}(\omega) \approx \frac{1}{2\pi} \left( \frac{\partial \varepsilon}{\partial \Phi} \right)^2 M^2 (2k_B T + 2\hbar\omega) / R. \quad (33)$$

This expected noise falls about 100 times below the measured PSD at  $f_{01} = \Delta/2\pi$ , indicating that intrinsic, microscopic noise mechanisms dominate the flux noise. With  $R$  unchanged, we would infer an effectively  $\sim 10$  times larger  $M$  to account for the observed noise.

Within this work, we did not identify which noise sources were responsible for the  $\delta\Delta$  noise, which can arise from, *e.g.*, critical-current noise or charge noise due to various microscopic mechanisms.

<sup>1</sup>Equation (33) is written with the following (“type 2”) definition of the Fourier transform, used throughout this paper:  $S_\lambda(\omega) = (1/2\pi) \int_{-\infty}^{\infty} dt \langle \lambda(0)\lambda(t) \rangle \exp(-i\omega t)$ , where  $\langle \cdot \rangle$  is the *un-symmetrized* correlation function.



Supplementary Figure 8:  $T_1$  **relaxation**. Repeated measurements of  $T_1$  relaxation from the excited state, at flux degeneracy  $\Phi_b = 0$ , showing how  $T_1$  varies by about 10 % from one measurement to the next. The size of the error bars from the exponential fits are on the order of the variation. Each trace took 1 min 40 s to measure;  $T_1$  may vary on a shorter time scale than that.

Taking the tilted quantization axis into account, so that the transverse noise is constituted by  $\delta\varepsilon$  noise at  $\varepsilon = 0$ , by  $\delta\Delta$  noise at  $\varepsilon \gg \Delta$ , and by combined projections of the two noises in-between, we arrive at the combined expression for the effective transverse Johnson–Nyquist noise,

$$S(\omega_{01}) = \frac{\Delta^2}{\omega_{01}^2} S_\varepsilon^{\text{JN}}(\omega_{01}) + \frac{\omega_{01}^2 - \Delta^2}{\omega_{01}^2} S_\Delta^{\text{JN}}(\omega_{01}). \quad (34)$$

As mentioned above, this *effective* noise is obviously greater than the expected extrinsic noise sources (Johnson–Nyquist as well as  $1/f$  noise). The subgap conductance, typically several hundred to several thousand times the normal resistance, may limit the achievable  $T_1$ -relaxation times [19, 20], a topic worth further studies.

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